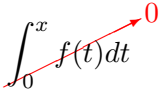


# Examples

Mudasir

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

let  $f(x) = \int_0^x f(t) dt = 1$



Derivatives:

$$\frac{dy}{dx}, \quad \frac{de^{x^2}}{dx} = 2xe^{x^2}$$

Partial derivatives:

$$\frac{\partial u}{\partial x}, \quad \frac{\partial^2 u}{\partial x^2}$$

Integrals:

$$\int_0^1 x^2 dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}$$

Differentials (explicit d):

$$\int f(t) dt$$

Vector calculus / del operator:

$$\vec{\nabla} f, \quad \vec{\nabla} \cdot \vec{F}$$

Laplace transform:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

Inline code example: `disp('Hello, world!')`

Block code example:

```
1 function z = f(t, y)
2     z = zeros(2,1);
3     z(1) = y(2);
4     z(2) = (-5*y(2)) + (-6*y(1));
5 end
6
7 tspan = [0 5];
8 y0 = [0, 1]; % initial condition y(0) = 0, dy/dt(0) = 1
9 [t, y] = ode45(@f, tspan, y0);
10
11 plot(t, y(:,1), '-o')
12 xlabel('Time t')
13 ylabel('y(t)')
14 title('Solution of ODE')
15 grid on
```

```
16 hold on
17 exact = -exp(-3 * t)+exp(-2 * t);
18 plot(t, exact, '-r')
```